

Online Test-Time Adaptation for Generalizable Dynamic Graph Anomaly Detection

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Abstract—Generalizable dynamic graph anomaly detection (DGAD) enables pretrained detectors to identify anomalies in unseen target domains without costly retraining. However, existing methods often fail for two reasons. First, they mainly rely on domain-agnostic patterns and miss domain-specific patterns that keep evolving. Second, they assume access to the full target domain data, whereas in more practical online test-time adaptation settings, target data arrive sequentially in unlabeled chunks. To address these limitations, we formulate online test-time adaptation for generalizable DGAD and propose OTTA-DGAD. OTTA-DGAD first extracts dynamic prototypes, i.e., evolving representations of normal and anomalous patterns, from temporal ego-graphs and stores them in a memory buffer. The buffer selectively retains general patterns shared across the source domains used for pretraining while incorporating new patterns from the target domain. An anomaly scorer then compares incoming edge representations against these prototypes to identify both general and domain-specific anomalies. During adaptation, OTTA-DGAD updates the memory buffer using reliable pseudo-labels identified through confidence-based detection. It further enriches each target chunk with relevant representations retained from previous chunks, compensating for information loss resulting from the sequential arrival of data. Extensive experiments under strict test-then-adapt OTTA settings demonstrate state-of-the-art performance on ten real-world datasets from diverse domains.

Index Terms—Online Test-time Adaptation, Dynamic Graph Anomaly Detection, Graph Neural Networks.

I. INTRODUCTION

GRAPH anomaly detection (GAD) aims to identify unusual nodes or edges from their features and structural relations. It has become increasingly important in applications such as fraud detection, transaction monitoring, and social-network security [1], [2], [3]. In practice, however, many graphs are evolving rather than static [4], [5], [6], [7]. For examples, new accounts being added with existing entities update their profiles, and interactions evolve over time. These making anomalous behaviors inherently dynamic rather than fixed. This motivates dynamic graph anomaly detection (DGAD), which capture not only abnormal patterns, but also how such abnormality evolves over time [8], [9], [10], [11], [12]. Existing DGAD [2], [13] often leverage transformers,

or unsupervised methods to detect anomalies by measuring deviations from normal patterns.

However, this one-model-per-domain [14], [15], [16] paradigm generalizes poorly due to following limitations. Firstly, new domains exhibit different anomaly semantics and feature, namely domain shift [17], [18], [19]. For example, a sudden increase in transactions involving a small group of accounts may indicate suspicious collusion in a transaction network. However, a similar surge in traffic flow around a small set of locations may simply be caused by the opening of a new shopping mall rather than anomalous behavior. As a result, a detector trained on one domain may fail to generalize to another. A straightforward solution is to retrain the detector for each new domain. Nevertheless, this is costly and difficult to scale, especially in real world where new domains continuously emerge and domain shift continuously happen [7], [4]. Moreover, target data arrive continuously in chunks rather than becoming available all at once [20], [21]. Waiting for the complete data stream delays timely adaptation. On the other hand, directly updating on each chunk will cause overfit to short term patterns since a single chunk only contains limited local observations.

These limitations motivate online test-time adaptation [22], [23], [24], [25] for generalizable DGAD. Specifically, a detector pretrained on a source domain can adapt to sequentially arriving unlabeled target chunks under continuous domain shift while maintaining anomaly detection accuracy.

Under this problem, the first major challenge is anomaly pattern, such as feature space, evolving over time between and within domain as shown in Fig. 1(a) [26], [17], [18], [19]. A pattern that appears abnormal at an early stage may later become common as the target domain evolves. This makes adaptation fundamentally difficult. Methods based on temporal ego-graphs [14] can capture evolving local anomaly patterns within a domain. However, these patterns are often closely tied to the domain's graph structure and anomaly semantics. Directly adapting them to a new domain may therefore lead to substantial mismatch and cause performance degradation. Recapturing temporal patterns for every target domain is also expensive and limits scalability. In contrast, cross-domain generalization methods retain domain agnostic pattern using prototypes [27], [28], [29], normalization, or prompts as shown in the upper part of Fig. 1(b). Although these mechanisms improve generalizability, they may fail to capture newly emerging, domain specific anomaly patterns in target domain. Therefore, model needs to preserve domain agnostic as well as specific patterns at the same time as shown

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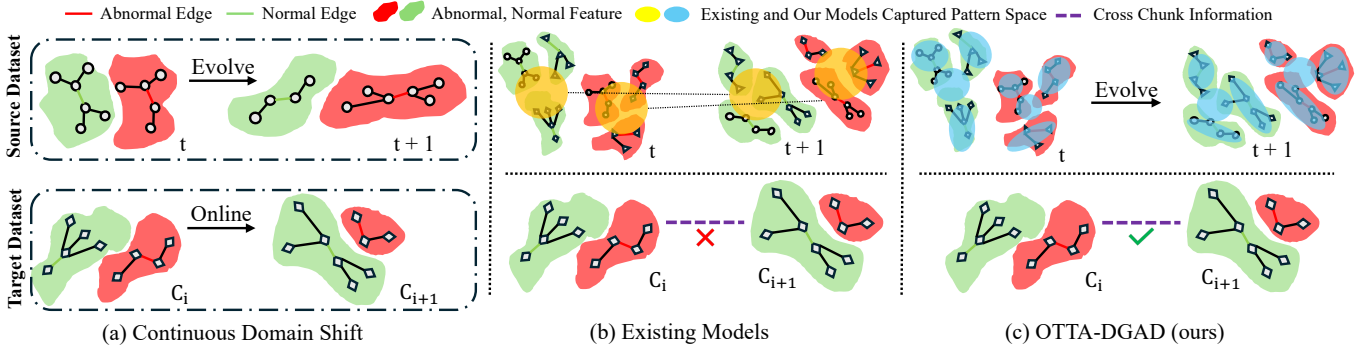


Fig. 1. Motivating experiments. (a) Anomalous patterns change over time, chunks and cross datasets. (b) Existing generalist methods capture only domain agnostic patterns, without cross chunk information preservation. (c) Our method, OTTA-DGAD, captures both domain-agnostic and domain-specific patterns. Cross chunk information is also preserved.

in the upper part of Fig. 1(c).

Secondly, chunk-wise and unlabeled nature of the online target data stream also introduce another challenge [30], [31], [32]. In dynamic graphs, each incoming chunk only provides a partial view of the target domain as shown in the lower part of Fig. 1(b). However, many anomaly patterns depend on cross chunk interactions, historical neighbors, and evolving structural dependencies. These information are not directly observable from the current chunk [33], [34]. For example, in a social network, a user may interact with only a few accounts within one time chunk. Notwithstanding, suspicious pattern may only become evident after linking multiple chunks. Consequently, chunk-wise adaptation fail to capture temporal evolution of graph structure and feature distribution, thus lead to incorrect anomaly detection. Although OTTA can retain cross chunk information by tracking historical scores or gradients, existing methods [35], [36] are mainly designed for sequential data. Moreover, they lack an explicit module for incorporating structural information across chunks. The problem is further complicated by the absence of labels [3], [37]. Therefore, the model needs to preserve cross chunk information as shown in lower part of Fig. 1(c).

To address the aforementioned challenges, we propose Online Test-Time Adapted Generalizable Dynamic Graph Anomaly Detector (OTTA-DGAD). OTTA-DGAD is pre-trained on multiple source-domain datasets to capture domain agnostic patterns. Then, it is adapted online to chunk-wise target data to incorporate target-specific patterns. Specifically, it first constructs dynamic prototypes, namely representations of normal and abnormal patterns, from spatial anomalies captured by temporal ego-graphs. These prototypes are further updated with edge-level temporal attributes, allowing them to reflect how anomalous patterns evolve over time. Prototypes collected from different source domains are stored in a memory buffer, which is then updated with prototypes from the new domain. Thus, the memory buffer preserves both domain-agnostic knowledge and domain-specific knowledge for adaptation. After that, OTTA-DGAD derives normal and abnormal distributions from the buffered prototypes for anomaly scoring.

A preliminary version of this work was published in [38]. Although DP-DGAD performed well, it had three major

limitations that prevent it from real world deployment. First, it cannot handle the missing information cross chunks, which we call context [39], caused by chunked target data stream. To address this issue, we preserve representations that are representative of the chunk distribution and distinct from one another in a context buffer. We measure the former using a coverage score and the latter using a difference score. These representations are later used to enhance the similar query representations in the next chunk. Moreover, DP-DGAD uses low-entropy detection pairs as confident predictions and uses their pseudo labels to update the prototypes. However, entropy alone measures only prediction certainty, not whether the prediction is structurally consistent with the graph. It can therefore select confidently incorrect samples. Moreover, selecting a fixed number of pseudo-labeled samples per class can bias prototype updates when target chunks are class-imbalanced, causing errors to accumulate over time. To address these issues, we assess pseudo-label quality using two complementary signals. The first is prediction entropy, which reflects model confidence. The second is structural support, which reflects whether a node’s predicted label agrees with those of its neighbors. We combine these signals into a reliability score, and normalize the score within each pseudo-label class to ensure comparability across classes. We then rank all detections together, regardless of their predicted classes, to select the most reliable ones. This allows the number of selected normal and anomalous detections to vary naturally with the confidence distribution of each chunk. Third, DP-DGAD has not been evaluated in an OTTA setting, leaving its effectiveness under realistic online test-time adaptation unclear. Therefore, in this work, we reformulate the task under the OTTA-DGAD setting. Unlike prior evaluations that assume fully available target data, we re-run all experiments under a strict chunk-wise online adaptation protocol, covering all baselines across all ten datasets.

- We propose a new and more realistic research problem, online test-time adaptation for generalizable dynamic graph anomaly detection. The pretrained DGAD model must adapt to unlabeled target dynamic graphs that arrive sequentially in chunks under continuous domain shifts.
- We equip OTTA-DGAD with two new components:

a structurally supported pseudo-labeling strategy and a cross-chunk information preserving mechanism. Together, they enable the detector to capture anomaly patterns that are both shared across chunks and specific to the current chunk under OTTA constraints.

- We conduct extensive experiments on a large-scale cross-domain benchmark containing ten real-world datasets from diverse domains. The results show that our method achieves state-of-the-art performance with improved stability and generalization.

II. RELATED WORK

A. Generalizable Dynamic Graph Anomaly Detection

Existing dynamic graph anomaly detection (DGAD) follow a general pipeline by modeling structural and temporal dependencies jointly. For example, TADDY [13] employs a transformer-based architecture to encode temporal evolution and spatial dependencies for anomaly detection. StrGNN and DynAnom [40], [41] focus on local subgraphs to efficiently identify structural irregularities in evolving graphs. In semi-supervised settings, methods such as SAD and CoLA [42], [13] exploit limited anomaly labels to learn discriminative representations [43]. Meanwhile, unsupervised DGAD methods [14] have received increasing attention because anomaly labels are often unavailable in practical applications. These methods generally assume that the training data are dominated by normal samples and detect anomalies according to their deviations from learned normal patterns. For instance, FALCON [15] exploits fine-grained temporal information through enhanced sampling and representation learning, while SLADE [16] models long-term evolving interactions to characterize normal behavioral patterns. Despite their effectiveness, most existing DGAD methods are developed under a closed-world, one-model-per-domain paradigm. They assume that the source and test graphs share similar feature distributions, structural patterns, and anomaly semantics. As a result, their performance can degrade substantially when deployed on unseen domains with distribution shifts [44], [45].

To alleviate cross-domain distribution shifts, recent studies have explored generalist detection models, which aims to learn general domain agnostic patterns from multiple source domains and apply it to unseen target domains [27], [28], [29], [46]. Existing approaches typically reduce domain discrepancies through unified representation strategies, including normalization [47], [48], prototype-based modeling [49], [33], [26], and prompt-based adaptation [50]. These methods seek to preserve domain-invariant patterns shared across graphs and thereby avoid retraining a dedicated detector for every target domain. Most generalist GAD methods are designed for static graphs and primarily preserve domain agnostic pattern, making them inadequate for evolving, domain specific anomaly patterns in unseen domains. Our method addresses this limitation by maintaining dynamic prototypes from temporal ego-graphs in a memory buffer, which retains general pattern while incorporating domain specific patterns.

B. Online Test Time Adaptation for Dynamic Graph Anomaly Detection

Online test-time adaptation (OTTA) [31], [22], [23] aims to adapt a source-pretrained model to an unlabeled target stream that arrives sequentially. Unlike conventional domain adaptation [7], [4], [51], [52], OTTA performs adaptation without accessing source data or target labels. Existing OTTA methods often update selected model parameters using self-supervised objectives on incoming target data. For example, CMF [53] filters momentum in the parameter space to stabilize continual adaptation, while TEA [21] performs adaptation by aligning target representations with class-wise energy distributions. SoTTA [20] improves robustness to noisy online streams by selecting reliable target samples and reducing the influence of noisy pseudo-labels. Protected TTA [32] further introduces online entropy matching and a statistical protection mechanism to avoid harmful self-training updates. Other methods use target-data selection, memory mechanisms, or diverse augmented views to improve adaptation stability. For instance, Universal TTA [35] combines weight ensembling, diversity-aware weighting, and prior correction to improve robustness across different test-time shifts. In addition, Dual Memory Networks [54] maintain complementary memory representations to support adaptation, whereas prompt-based approaches adapt a small set of prompts or lightweight modules while retaining the pretrained backbone [55], [24]. These methods improve adaptation efficiency and mitigate domain shift. However, they lack module for dynamic graphs that both need temporal as well spatial adaptation.

For dynamic graphs, recent studies have begun to investigate test-time adaptation through lightweight prompt tuning and continual adaptation. PromptDyG [25] performs test-time prompt adaptation on dynamic graphs by keeping the pretrained backbone fixed and updating lightweight prompts to accommodate evolving patterns. ADCSD [56] retains a frozen source-pretrained backbone and refines its predictions through a short-term correction module conditioned on the current chunk's statistics and a long-term module based on an exponential-moving-average historical state. Followed by LCoTTA [36] that maintain a queue of recent adaptation gradients, estimates their principal subspace online, and projects each current update onto this subspace before updating lightweight adaptation parameters. Notwithstanding, these methods do not explicitly preserve or retrieve cross chunk information that capture evolving structural dependencies and can suffer from suboptimal performance when required to identify anomalies in dynamic graph streams.

III. PROBLEM FORMULATION

Definition 1. (Dynamic Graph Anomaly Detection) A dynamic graph can be denoted as $\mathcal{G} = \{G_1, G_2, \dots, G_T\}$, where T is the number of intervals, and interval t consists of M_t timestamps. Specifically, $G_t = (V_t, E_t)$, where V_t is a node set and E_t is an edge set, with $A^{V_t} \in \mathbb{R}^{N^{V_t} \times N^{V_t}}$ and $A^{E_t} \in \mathbb{R}^{N^{E_t} \times N^{E_t}}$ denoting the node and edge adjacency matrices ($N^{V_t} = |V_t|$, $N^{E_t} = |E_t|$), and $X^{V_t} \in \mathbb{R}^{N^{V_t} \times D^V \times M_t}$, $X^{E_t} \in \mathbb{R}^{N^{E_t} \times D^E \times M_t}$ denoting the temporal node and edge

features, D^E, D^V is the dimension of features, respectively. Each edge is associated with a label vector $y_t \in \{0, 1\}^{N^{E_t}}$, partitioning E_t into the normal edge set E_t^n and the abnormal edge set E_t^a .

Following existing online test-time adaptation methods [23], we further divide $\mathcal{G}$ into Q data chunks $\{\mathcal{C}_1, \mathcal{C}_2, \dots, \mathcal{C}_Q\}$ for online processing. Each chunk $\mathcal{C}_q = (V_q, E_q)$, aggregates M_q consecutive timestamps and contains a fixed number of edges $|E_q| = B$ (e.g., $B = 128$ interactions). $\mathcal{C}_q$ inherits the same notation as $\mathcal{G}_t$ (i.e., $A^{V_q}, A^{E_q}, X^{V_q}, X^{E_q}, E_q^n, E_q^a, y_q$). The goal of dynamic graph anomaly detection is to distinguish abnormal edges from normal ones based on the evolving graph structure and temporal features.

Definition 2. (Continuous Domain Shift) Given $\{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{S+K}\}$ being a collection of $(S + K)$ dynamic graphs from different domains, where $\{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_S\}$ are from source domain and $\{\mathcal{G}_{S+1}, \mathcal{G}_{S+2}, \dots, \mathcal{G}_{S+K}\}$ are from target domains. For two dynamic graphs from different domains, $\mathcal{G}_i$ and $\mathcal{G}_j$ where $i \neq j$, we define domain shift as a difference between their distributions of node and edge features, graph structures, such as the adjacency matrices or their mappings from edges to anomaly labels. Such domain shifts occur among any two dynamic graphs in the dataset collection $\{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{S+K}\}$.

Problem Definition. (Online Test-Time Adaptation for Generalizable Dynamic Graph Anomaly Detection) Given S source dynamic graphs $\{\mathcal{G}_1, \dots, \mathcal{G}_S\}$, we first train a source detector Ψ_0 that distinguishes abnormal from normal edges on source dynamic graphs. Given an unseen target dynamic graph $\mathcal{G}_k$ partitioned into Q_k chunks $\{\mathcal{C}_1, \dots, \mathcal{C}_{Q_k}\}$, our goal is to sequentially adapt Ψ_0 into a series of detectors $\{\Psi_1, \dots, \Psi_{Q_k}\}$ that minimize the anomaly detection error across all chunks of $\mathcal{G}_k$, without access to any target label. Specifically, for each chunk $\mathcal{C}_q$, the detector Ψ_{q-1} adapted via previous chunk $\mathcal{C}_{q-1}$ first predicts anomaly scores on the current unlabeled chunk $\mathcal{C}_q$ as:

$$\hat{y}_q = \Psi_{q-1}(A^{V_q}, A^{E_q}, X^{V_q}, X^{E_q}), \quad (1)$$

and is then updated to Ψ_q using only $\mathcal{C}_q$ and $\hat{y}_q$, without access to y_q or future chunks.

IV. METHODOLOGY

As shown in Fig. 2, OTTA-DGAD first extracts evolving patterns using dynamic prototype extraction and stores them in a memory buffer. Then, during cross-domain anomaly scoring, it retains general patterns shared among pretrained source datasets while incorporating domain-specific ones. These dynamic prototypes are compared with embeddings to calculate anomaly scores. On target data, it measure detection with a reliability score that combines prediction entropy and structural support, where structural support reflects agreement between a node and its neighbors. These detections provide pseudo-labels for updating the memory buffer. Besides, a context buffer stores representations is used to enrich cross chunk information. We select samples using two scores. The coverage score measures similarity to the distribution of the current chunk, while the difference score measures dissimilarity from

entries already stored in the buffer. When a new chunk arrives, its query representations is enriched by cross chunk information from this buffer.

A. Dynamic Prototype Extraction

In this section, our goal is to extract dynamic prototypes as the evolving representations of normal and anomalous patterns. Particularly, we align prototypes with both abnormal and normal patterns to fully exploit anomaly discriminability. This approach enables a more effective capture of evolving anomalous patterns over time.

For each edge e_i , its anomalous pattern can be captured by the ego-graph consisting of its neighbors, thereby reflecting how it deviates from or resembles them [28]. To capture not only the anomalous pattern but also its evolution over time, temporal ego-graphs are utilized. These graphs [14] extract k -hop neighboring edges occurring on or before timestamp t around edge e_i and form a subgraph G_{ego} . The temporal ego-graph is then input into a simple backbone model consisting of a Graph Neural Network (GNN), followed by a transformer to retrieve representations. The GNN first outputs representation $H_l \in \mathbb{R}^{|E_{ego}| \times d}$ as follows:

$$H_l = \sigma(A_{ego}H_{l-1}W_1^l + H_{l-1}W_2^l), \quad (2)$$

where d is the dimension of representation and l is the GNN layer index. $W_2^l, W_1^l \in \mathbb{R}^{D^{l-1} \times D^l}$ are learnable parameter matrix of layer l and σ is the activation function. A_{ego} refers to the edge adjacency matrix of the ego-graph. To encourage the representation to capture more domain-agnostic features, a residual module is applied here to smooth edge-specific representation $h_i \in H_l$ as follows:

$$h'_i = h_i - \frac{1}{|E_{ego}|} \sum_{e_j \in E_{ego}} h_j. \quad (3)$$

Then, the processed H'_l is fed into a transformer. This transformer's role is to identify the similarity between edge e_i and its k -hop neighbors. In this way, we encode the spatial anomalous patterns of e_i as they evolve over time into its embedding z_i .

$$z_i = \sum_{h'_j \in H'_l} \frac{\exp(\frac{\langle w_Q h'_i, w_K h'_j \rangle}{\sqrt{d_{out}}})}{\sum_{h'_j \in H'_l} \exp(\frac{\langle w_Q h'_i, w_K h'_j \rangle}{\sqrt{d_{out}}})} w_V h'_j, \quad (4)$$

where $\langle \cdot, \cdot \rangle$ denotes the dot product, d_{out} refers to the dimension of the output representation z_i of edge e_i . w_Q, w_K, w_V refer to the Query, Key, and Value matrix.

After obtaining the representation of each edge, we align dynamic prototypes with both the abnormal and normal edges' representations. Formally, we have $p_n \in \mathbb{R}^{d_p}$ as the dynamic normal prototypes while $p_a \in \mathbb{R}^{d_p}$ represents the dynamic abnormal prototypes. Both for the abnormal e_i in E_t^a and that in E_t^n , we can obtain the ego-graph representation. We denote the representation of abnormal edges as Z_a and the

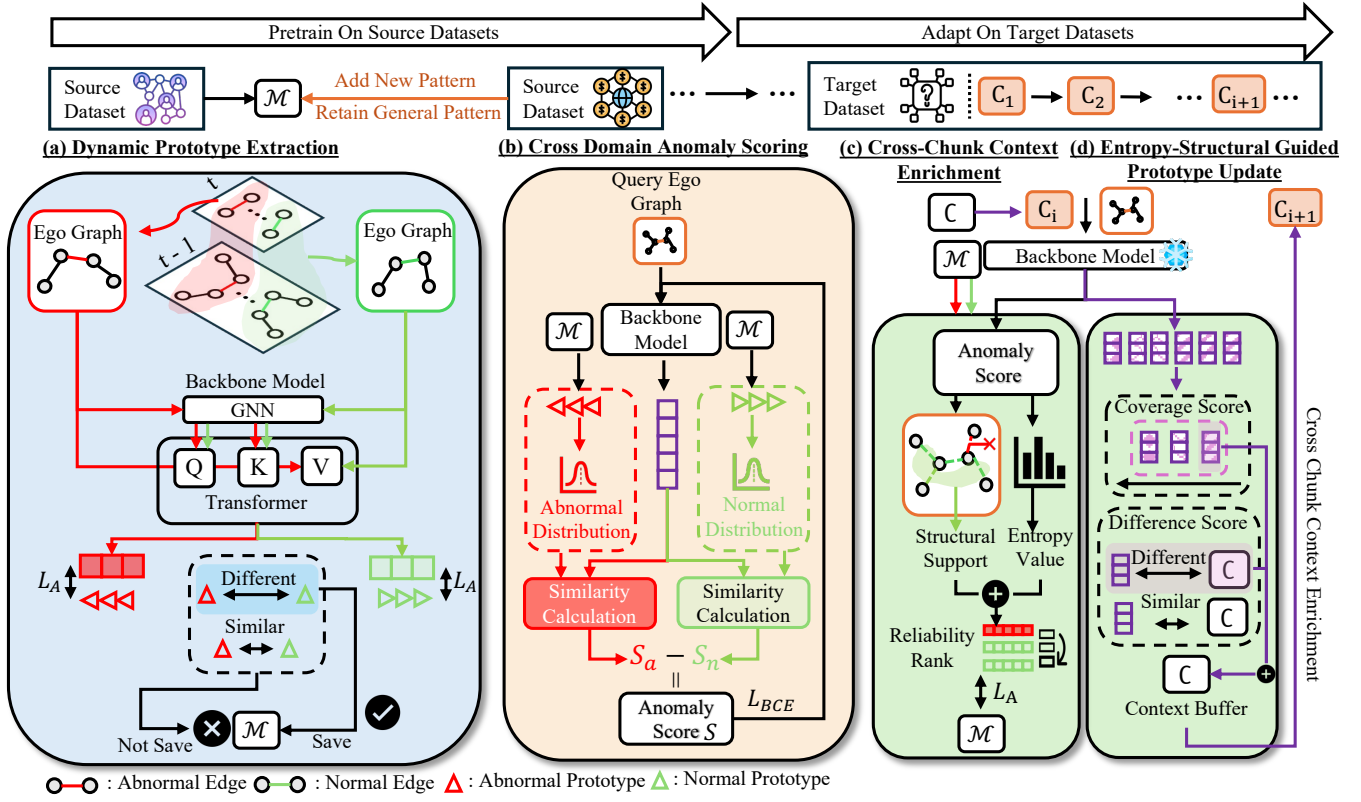


Fig. 2. **General framework of OTTA-DGAD.** (a) Starting with the first source dataset, we extract ego-graphs to capture temporal patterns and store the most distinct prototype pairs in a memory buffer. (b) As we pretrain on the following source datasets, new domain-specific patterns are added while general domain-agnostic patterns are retained. Prototype distributions are then compared with edge embeddings for anomaly scoring. (c) On target chunks, reliable detections, measured by prediction entropy and agreement with neighboring edges, provide pseudo-labels for memory updates. The buffer retains representations with high similarity to chunk distribution, which we define as coverage score and different with stored entries. These representations enrich subsequent chunks with cross-chunk context, while the generalist model remains frozen.

representations of normal edges as Z_n . To align p_a and p_n with Z_a and Z_n , we define alignment loss function as follows:

$$L_A = \sum_{i=1}^{|E_t|} I_{y_i=0} \|z_i - p_n\|_2^2 + I_{y_i=1} \|z_i - p_a\|_2^2, \quad (5)$$

where $I_{y_i=0}$ represents indicator function. This function returns 1 when the condition $y_i = 0$ is met, and 0 otherwise.

These aligned dynamic prototypes are further stored in the dynamic prototype buffer $\mathcal{B}$ that has size $\mathcal{M}$, set as 10% of the training source dataset size. The buffer is updated during each iteration to capture diverse aspects of the abnormal and normal patterns as possible. Due to limitations in size and memory, we retain only the most discriminative prototypes. This is achieved by ranking all prototypes currently in the buffer in ascending order based on the distance score. The prototype pairs with the smallest distance score, indicating weak separability between normal and abnormal patterns are therefore replaced. The distance score is calculated as the mean Euclidean distance between pairs of prototypes:

$$s_d = \frac{1}{d_p} \sum_{m=1}^{d_p} \|p_a^m - p_n^m\|_2, \quad (6)$$

where d_p refer to the dimension of prototypes, p_a^m, p_n^m refer to the corresponding dimension m 's feature in prototypes. In each training epoch, the system identifies the pair of prototypes that exhibits the highest distance score. This most distinct pair is then used to set the initial values for p_n and p_a for that specific epoch. This strategy ensures that the dynamic prototypes are continuously refined to be more distinguishable.

B. Cross Domain Anomaly Scoring

Distribution-based anomaly scoring surpasses traditional binary classifiers in its ability to generalize across different domains. It captures higher-order statistics of data distributions by comparing data with both abnormal and normal distributions. However, our current dynamic prototypes are domain-specific, which prevents their direct application across different domains. To generalize them to new domains, we identify and save domain-agnostic ones, while adding new prototypes from new domains. Subsequently, we build anomaly scoring module to leverage these enhanced prototypes.

For identifying domain-agnostic patterns, they should already be present in the memory buffer, which stores information from previously encountered source datasets. Additionally, they should also exhibit similarity to the feature representations found in the subsequent source dataset. The similarity s_e is calculated as the mean Euclidean distance

between the dynamic prototype pair and the representations of edges in the new domain dataset:

$$s_e = \frac{1}{|E_t|} \sum_{i=1}^{|E_t|} \|z_i - p_n\|_2 + \|z_i - p_a\|_2. \quad (7)$$

Notably, the representation z_i is projected to the same dimension as that of the dynamic prototypes. To ensure s_d and s_e are on a comparable scale before combination, s_e is likewise normalized via d_p . A lower distance indicates higher similarity, meaning more general prototypes. This will be combined with s_d in the following manner:

$$s_r = \lambda_d s_d - \lambda_e s_e, \quad (8)$$

where λ_d and λ_e are parameters used to control the contribution of the two scores. A higher s_r indicates greater pairwise distance and similarity to the new source dataset, representing more general domain-agnostic patterns. For new domain patterns, we rank buffer prototypes by s_r in ascending order and replace the lowest-scoring pair with one from the new domain. We then initialize p_n and p_a using the highest s_r pairs to ensure dynamic prototypes capture both domain-agnostic and domain-specific patterns.

With these dynamic prototypes in the memory buffer, we build an anomaly scoring module by measuring edge representation similarity to both normal and abnormal distributions. Top prototype pairs alone cannot fully represent distributions, so we iteratively update statistical measures (mean and covariance). Each iteration incorporates new data and existing statistics to capture evolving distributions. The mean update is as follows:

$$\mu_{n,t} = \alpha \mu_{n,t-1} + (1 - \alpha) \sum_{i=1}^{\mathcal{M}} p_{n,i}, \quad (9)$$

$$\mu_{a,t} = \alpha \mu_{a,t-1} + (1 - \alpha) \sum_{i=1}^{\mathcal{M}} p_{a,i}, \quad (10)$$

where μ_n and μ_a are the mean values for normal and abnormal distributions. α is a momentum parameter to control the ratio of updating based on prototypes and the existing mean value.

For covariance updates, we first calculate center embedding, which captures how individual prototypes differ from the overall average. Then, we use center embeddings to update the covariance that inherently measures how data points vary together around their mean values:

$$C_n = [p_{n,1} - \mu_{n,t}, p_{n,2} - \mu_{n,t}, \dots, p_{n,\mathcal{M}} - \mu_{n,t}]^T, \quad (11)$$

$$C_a = [p_{a,1} - \mu_{a,t}, p_{a,2} - \mu_{a,t}, \dots, p_{a,\mathcal{M}} - \mu_{a,t}]^T, \quad (12)$$

$$\Sigma_{n,t} = \alpha \Sigma_{n,t-1} + (1 - \alpha) \frac{C_n^T C_n}{\max(1, \mathcal{M} - 1)}, \quad (13)$$

$$\Sigma_{a,t} = \alpha \Sigma_{a,t-1} + (1 - \alpha) \frac{C_a^T C_a}{\max(1, \mathcal{M} - 1)}, \quad (14)$$

where C_n and C_a are the center embeddings, and, Σ_n and Σ_a are the covariance values.

Then we compute a learned discriminant score between the edge representation z_i and each distribution using a bilinear form parameterized by μ and Σ :

$$s_{n,i} = z_i^T \mu_n - \lambda_n z_i^T \Sigma_n z_i, \quad (15)$$

$$s_{a,i} = z_i^T \mu_a - \lambda_a z_i^T \Sigma_a z_i, \quad (16)$$

where λ_a, λ_n are learnable parameters, adjusting the relative contribution of the second-order term, that are learned during source pretraining and kept fixed throughout target-time adaptation. Next, we compute the anomaly score $s_i = s_{a,i} - s_{n,i}$, which measures the similarity gap to determine if an edge is closer to the abnormal or normal distribution. This score is then converted into a probability (p_i) using a sigmoid function, representing the likelihood of the edge being anomalous. The overall learning objective combines binary cross-entropy loss with the previous alignment loss (L_A):

$$L_{BCE} = -\frac{1}{|E_t|} \sum_{i=1}^{|E_t|} y_i \log(p_i) + (1 - y_i) \log(1 - p_i), \quad (17)$$

$$L = \lambda_{BCE} L_{BCE} + \lambda_A L_A, \quad (18)$$

where λ_A and λ_{BCE} are parameters controlling the contribution ratio of these two losses to the final loss.

C. Entropy-Structural Guided Prototype Update

To adapt the source-pretrained detector to an unlabeled target stream, we update the dynamic prototype memory using only reliable pseudo-labeled target edges. Unlike DP-DGAD [38], which relies solely on entropy-ranked confident detections, we add structural support as an additional criterion, and then perform class-wise memory updates to alleviate the bias caused by imbalanced confident pseudo-labels.

Given an incoming target chunk $\mathcal{C}_q$, the pretrained anomaly scorer produces an anomaly probability $p_i \in [0, 1]$ for each target edge e_i based on the enhanced representation $\tilde{z}_i$, together with its pseudo-label

$$\hat{y}_i = \mathbb{I}(p_i > 0.5), \quad (19)$$

where $\hat{y}_i = 1$ and $\hat{y}_i = 0$ denote abnormal and normal pseudo-labels, respectively.

We first estimate the prediction confidence using entropy:

$$\mathcal{H}(p_i) = -p_i \log(p_i + \epsilon) - (1 - p_i) \log(1 - p_i + \epsilon), \quad (20)$$

where lower entropy indicates higher prediction confidence. We further convert it into a normalized confidence score:

$$ec_i = 1 - \frac{\mathcal{H}(p_i)}{\ln 2}. \quad (21)$$

However, confidence alone is insufficient under continuous domain shift. We therefore introduce a structural support score to evaluate whether the predicted class of e_i is supported by its local neighborhood. Let $\mathcal{N}_i$ denote the neighboring edges of e_i in its temporal ego-graph. The structural support score is defined as

$$ss_i = \begin{cases} \sum_{j \in \mathcal{N}_i} (1 - p_j), & \hat{y}_i = 0, \\ \sum_{j \in \mathcal{N}_i} p_j, & \hat{y}_i = 1, \end{cases} \quad (22)$$

In this way, a pseudo-labeled normal edge is supported by normal-like neighbors, while a pseudo-labeled abnormal edge is supported by abnormal-like neighbors. Based on the above two criteria, we assign each target edge a raw reliability score:

$$r_i = ec_i \cdot ss_i. \quad (23)$$

Since the structural support score is computed in a class-conditional manner, the raw reliability scores of pseudo-labeled normal and abnormal edges may be on different scales. To make the scores comparable for global ranking, we further normalize them within each pseudo-label class. Specifically, let:

$$r_{\min}^{(y)} = \min_{\{i|\hat{y}_i=y\}} r_i, \quad r_{\max}^{(y)} = \max_{\{i|\hat{y}_i=y\}} r_i, \quad (24)$$

where $y \in \{0, 1\}$. The class-normalized reliability score is defined as

$$\tilde{r}_i = \frac{r_i - r_{\min}^{(\hat{y}_i)}}{r_{\max}^{(\hat{y}_i)} - r_{\min}^{(\hat{y}_i)} + \epsilon}. \quad (25)$$

We then select the top N_{con} target edges with the highest normalized reliability scores as confident detections. According to their pseudo-labels, the selected confident detections are further divided into abnormal and normal subsets E_q^a, E_q^n . Using the selected confident detections, we construct class wise confident detection for the current chunk by normalized reliability weighted aggregation:

$$\mathbf{z}_q^a = \frac{\sum_{e_i \in E_q^a} \tilde{r}_i \tilde{\mathbf{z}}_i}{\sum_{e_i \in E_q^a} \tilde{r}_i + \epsilon}, \quad \mathbf{z}_q^n = \frac{\sum_{e_i \in E_q^n} \tilde{r}_i \tilde{\mathbf{z}}_i}{\sum_{e_i \in E_q^n} \tilde{r}_i + \epsilon}. \quad (26)$$

We then align the classwise confident detections with the corresponding memory prototypes through the alignment loss $\mathcal{L}_A$. Thus, only reliable pseudo-labeled target edges contribute to memory adaptation. By normalizing the reliability scores within each pseudo-label class before global ranking, the proposed strategy reduces the scale bias across classes while still allowing the numbers of selected abnormal and normal detections to adapt to the confidence distribution of each target chunk.

D. Cross Chunk Context Enrichment

Temporal ego-graph of a target edge, which contain temporal and structural context, may be incomplete within a single chunk. As a result, the pretrained detector may generate incorrect pseudo labels because of insufficient context. To address this issue, we introduce a Cross Chunk Context Enrichment module, which supplements the current target chunk with

historical context distilled from recent chunks before reliable pseudo-label selection and memory adaptation.

To remain consistent with the online test-time adaptation setting, we do not replay or reprocess full historical target chunks. Instead, we store only representation of ego graphs. Specifically, after processing chunk $\mathcal{C}_{q-1}$, we retain edges whose ego-graph representations representative of current chunk while remaining dissimilar to representations already stored in the buffer $\mathcal{B}_c$. Here, representative is measured by the extent of a certain representation similarity to other representations in the chunk. We denote this measurement coverage score and is defined as:

$$cs_i = \frac{1}{|E_{q-1}| - 1} \sum_{\substack{e_j \in E_{q-1} \\ j \neq i}} \text{sim}(\mathbf{z}_i, \mathbf{z}_j), \quad (27)$$

where $\text{sim}(\cdot, \cdot)$ denotes a similarity function, such as cosine similarity. The difference score that measure the dissimilarity to representations that already in buffer is defined as:

$$ds_i = 1 - \max_{\mathbf{z}_j \in \mathcal{B}_c} \text{sim}(\mathbf{z}_i, \mathbf{z}_j). \quad (28)$$

Then we multiply two score, also the temporal component being $\frac{q_i}{q-1}$, where q_i being the chunk id of specific representation. To ensure that the context buffer only contains only the most recent, representative, distinct, and informative representations, we compare each normalized multiplication score against the scores of the entries already in $\mathcal{B}_c$ and retain the top $\mathcal{M}_c$ entries. The capacity grows online by 10% of each incoming chunk's size, up to a fixed maximum capacity $\mathcal{M}_{c,\max}$.

Given a target edge e_i in chunk $\mathcal{C}_q$ with representation $\mathbf{z}_i$, we retrieve its cross chunk context from the context buffer by identifying stored representation that are semantically similar to the current edge. For each stored context representation $\mathbf{z}_j \in \mathcal{B}_c$, we compute a cross chunk similarity score using above like similarity function. Top N_{con} representation with the highest similarity scores as the historical context of e_i . Based on these retrieved context representation, we construct a cross chunk context vector $\mathbf{c}_i$ by weighted aggregation via aforementioned temporal component. The $\mathbf{c}_i$ is then fused with the current edge representation to obtain a context-enhanced target representation:

$$\tilde{\mathbf{z}}_i = \lambda_c \mathbf{z}_i + (1 - \lambda_c) \mathbf{c}_i, \quad (29)$$

where λ_c is parameter controlling the proportion of enrichment. For the first target chunk, the historical context buffer is empty. Therefore, no cross chunk context is retrieved and the enhanced representation degenerates to the current representation. The enhanced representation $\tilde{\mathbf{z}}_i$ is subsequently used during the target adaptation stage.

V. EXPERIMENTS

A. Datasets

We use 10 real-world datasets [13], [16], [57] from various domains for evaluation, and the detailed statistics of datasets are presented in Table I. For a dataset with no ground truth

anomalies. We follow previous work [13] to inject anomalies in random timestamps of the dataset. Specifically, take p to be the proportion of the total m number of samples in the dataset. We link $p \times m$ number of original disconnected nodes to be the injected anomalies.

TABLE I
STATISTICS OF DATASETS

Dataset	Domain	#Nodes	#Edges	Avg. Degree
Wikipedia	Online Collaboration	9,227	157,474	34.13
Bitcoin-OTC	Transaction Network	5,881	35,588	12.10
Bitcoin-Alpha	Transaction Network	3,777	24,173	12.80
Email-DNC	Online Communication	1,866	39,264	42.08
UCI Messages	Online Communication	1,899	13,838	14.57
AS-Topology	Autonomous Systems	34,761	171,420	9.86
MOOC	Online Courses	7,074	411,749	116.41
Synthetic-Hijack	Email Spam	986	333,200	38.11
TAX51	Transaction Network	132,524	467,279	7.05
DBLP	Citation Network	25387	119899	9.44

B. Performance Metrics and Baselines

We adopt AUROC and AUPRC as the evaluation metrics [13], [28], [48]. Higher metric value indicates better detection performance.

Since AUROC and AUPRC are undefined when a chunk contains only one class, we exclude such chunks from chunk-level metric computation. Let $\mathcal{K}_{valid}$ denote the set of valid target chunks that contain at least one normal edge and one abnormal edge. The final online evaluation results are obtained by averaging the chunk-level metrics over all valid chunks:

$$AUROC_{avg} = \frac{1}{|\mathcal{K}_{valid}|} \sum_{k \in \mathcal{K}_{valid}} AUROC_k, \quad (30)$$

$$AUPRC_{avg} = \frac{1}{|\mathcal{K}_{valid}|} \sum_{k \in \mathcal{K}_{valid}} AUPRC_k. \quad (31)$$

This protocol evaluates the model in a strictly online manner, where adaptation and prediction are both performed sequentially along the target stream without revisiting full historical target chunks. In this way, the reported results reflect not only the anomaly discrimination capability of the model, but also its ability to continuously adapt to evolving target distributions under the OTTA setting. We select following baseline methods for comparison:

- **DP-DGAD** [38]: DP-DGAD is pretrained on labeled source dynamic graphs to learn normal and anomalous prototypes, then adapts to unlabeled target streams by updating its prototype memory with high-confidence pseudo-labels. As it supports online target adaptation, we include it as an OTTA baseline.
- **GeneralDyG** [14]: For fair comparison, we extend the GeneralDyG with a transformer module and a binary classification head, enabling it to learn discriminative temporal ego-graph representations from labeled normal and anomalous source edges. During online target adaptation, it updates target-side prototypes using only reliable pseudo-labeled edges.
- **FALCON** [15]: We extend the FALCON with label-aware supervision and a BCE loss, enabling it to learn

discriminative temporal representations from normal and anomalous source edges. During online target adaptation, FALCON incrementally refines its temporal representation space using reliable pseudo-labeled target edges for anomaly scoring.

- **SLADE** [16]: We enhance SLADE with a binary classification objective to learn discriminative representations of normal and anomalous dynamic patterns from source data. During online target adaptation, it updates its temporal self-supervised objective using reliable pseudo-labeled target edges.
- **SAD** [42]: We combine SAD's deviation loss with label-aware contrastive learning to separate normal and anomalous temporal structures on source data. On target dataset, reliable pseudo-labeled edges are used to update its memory bank and refine target-side representations.
- **TADDY** [13]: Following the core idea of TADDY, we use a GNN encoder and a transformer-based temporal encoder to represent each dynamic graph chunk. The model is trained on labeled source edges with BCE loss and adapted online using reliable pseudo-label.
- **ARC** [27]: We extend ARC with a transformer-based temporal encoder to model dynamic graphs. It learns anomaly-aware representations from labeled source temporal neighborhoods and is adapted online using reliable pseudo-labeled target edges.
- **AnomalyGFM** [28]: We extend AnomalyGFM with a transformer-based temporal encoder to learn graph-agnostic normal and anomalous prototypes from labeled source data. During online target adaptation, the source encoder and abnormal prototype are fixed, while the normal prototype is incrementally refined using reliable pseudo-labeled target edges.
- **UNPrompt** [29]: We replace UNPrompt's backbone with a GNN-transformer encoder to model temporal subgraphs in dynamic graphs. It is pretrained on source data with dynamic graph contrastive learning and adapted online by updating neighborhood prompts using reliable pseudo-labeled target edges.
- **GraphPrompt** [50]: We replace GraphPrompt's backbone with a GNN-transformer encoder for temporal subgraph modeling. It aligns source representations with normal and anomalous prototypes and incrementally refines the prompt using reliable pseudo-labeled target edges during online adaptation.
- **ADCSD** [56]: We freeze the source-pretrained OTTA-DGAD backbone and replace cross chunk recovery and prototype evolution with lightweight short- and long-term score-correction modules. These modules are updated online using confident pseudo-labels to refine anomaly scores, while no cross chunk information or prototype memory is maintained.
- **LCoTTA** [36]: We construct LCoTTA-DGAD by freezing the source-pretrained DGAD backbone and adapting only lightweight parameters. For each target chunk, reliable pseudo-labels are used to compute adaptation gradients, which are projected onto the online tracked principal subspace of recent gradients before updating

TABLE II
OVERALL MODEL PERFORMANCE UNDER THE OTTA SETTING ACROSS EIGHT DATASETS WITH THREE ANOMALY PERCENTAGES.

Metrics	Category	Methods	AS-Topology			Bitcoin-Alpha			Email-DNC			UCI Messages			Bitcoin-OTC			TAX51			DBLP			Synthetic-Hijack		
			1%	5%	10%	1%	5%	10%	1%	5%	10%	1%	5%	10%	1%	5%	10%	1%	5%	10%	1%	5%	10%	1%	5%	10%
AUROC	DGADs	SAD	40.16	43.81	43.43	57.61	55.59	60.27	50.05	54.54	52.28	56.91	58.28	59.74	70.04	69.19	74.12	42.52	43.56	46.39	43.02	42.98	47.87	59.85	59.67	58.14
		SLADE	40.47	42.21	44.42	60.62	59.25	62.98	47.98	50.37	52.11	54.64	55.28	58.32	67.34	66.35	70.44	46.68	48.04	49.33	42.11	41.17	41.35	60.53	60.34	59.99
		FALCON	46.73	50.24	53.09	57.13	61.48	66.76	76.81	72.71	75.02	59.36	57.05	62.22	67.37	68.43	70.16	45.27	48.79	50.28	44.41	46.30	47.11	59.06	60.17	65.55
		GeneralDyG	41.99	43.51	45.80	62.26	63.03	65.09	50.93	52.63	54.78	50.33	53.50	58.91	70.91	71.77	73.84	52.79	50.58	50.42	48.33	47.34	46.78	56.78	53.47	54.02
		TADDY	40.11	42.57	41.08	43.69	45.44	42.03	50.77	53.27	56.90	49.82	50.39	56.12	44.47	50.79	48.72	40.48	42.21	46.94	40.32	40.59	42.40	58.77	60.59	62.10
	Generalists	GraphPrompt	48.23	50.66	52.32	46.08	46.24	43.48	67.59	70.11	72.31	59.93	58.29	58.74	48.36	50.07	52.25	51.24	53.33	56.22	58.62	57.15	61.35	68.58	74.42	70.96
		UNPrompt	50.29	52.21	56.57	43.01	46.73	48.62	68.34	71.07	72.58	56.74	58.28	60.01	43.09	50.35	52.85	53.77	54.61	58.05	60.37	56.69	62.20	70.32	73.64	77.58
		AnomalyGFM	52.13	53.90	57.84	46.23	47.01	49.56	70.62	72.39	74.16	60.03	62.44	59.59	52.80	51.22	50.45	54.97	55.71	56.87	60.24	57.83	59.27	72.52	78.07	76.35
		ARC	53.68	54.13	60.10	40.59	42.62	43.51	70.52	75.19	72.33	58.05	60.40	60.21	47.75	52.82	48.19	47.73	51.01	58.22	60.97	57.59	62.32	73.06	72.14	75.61
	OTTA	ADCSD	70.03	71.27	70.81	74.68	73.22	72.95	71.60	73.75	72.17	54.43	55.36	59.86	72.75	70.75	76.84	61.54	58.67	56.06	54.83	55.35	56.74	72.43	72.02	73.96
		LCoTTA	69.47	71.13	72.21	73.33	71.65	71.32	69.66	73.84	72.21	54.27	55.15	59.85	70.38	69.41	68.48	61.20	60.25	57.23	55.92	55.50	57.32	72.36	71.97	70.66
		DP-DGAD (OTTA)	69.48	71.04	71.99	62.51	65.34	69.77	70.21	74.42	71.09	59.86	55.22	59.69	76.74	71.30	68.59	55.68	60.33	57.62	60.41	54.30	57.66	70.81	70.62	69.36
		OTTA-DGAD	73.48	74.04	76.99	77.88	75.57	75.47	79.88	84.12	81.77	63.51	64.99	69.75	80.94	79.85	78.61	65.90	65.72	66.06	62.05	59.92	63.25	81.33	81.10	83.72
AUPRC	DGADs	SAD	5.26	11.15	11.87	5.12	7.45	10.16	10.88	13.16	25.96	10.47	14.75	24.85	6.91	11.42	17.30	4.76	7.03	13.58	3.62	6.84	9.02	9.72	14.89	16.66
		SLADE	5.34	10.26	11.52	5.49	6.05	10.97	10.95	18.06	23.23	10.47	16.72	23.19	5.57	9.11	12.02	4.75	6.02	11.71	2.61	5.79	10.01	10.68	15.05	19.50
		FALCON	7.01	11.37	11.91	5.99	7.13	12.87	14.39	20.06	32.07	12.85	20.39	27.79	6.06	10.77	13.14	9.23	8.19	14.20	4.44	7.56	8.47	12.38	16.13	20.52
		GeneralDyG	5.40	12.30	19.88	5.45	4.97	10.07	11.46	13.71	29.06	12.63	18.10	20.98	7.81	12.08	17.61	4.66	5.04	10.10	3.30	6.54	9.55	10.21	17.99	26.61
		TADDY	5.09	10.60	11.17	4.38	7.03	10.55	11.48	14.39	26.56	10.55	16.37	21.25	5.09	10.17	11.36	5.11	8.31	10.08	3.05	7.21	9.02	8.53	14.22	21.41
	Generalists	GraphPrompt	10.39	11.61	19.10	7.46	10.07	11.11	18.18	24.32	35.26	17.06	21.27	26.72	5.10	10.94	15.52	9.97	11.02	12.18	6.58	8.18	10.03	10.25	14.33	25.85
		UNPrompt	10.54	12.77	19.26	8.01	9.27	11.81	20.02	23.94	37.32	17.65	23.87	29.09	5.01	12.27	16.74	10.66	11.83	13.18	7.14	7.60	10.96	11.57	14.03	26.22
		AnomalyGFM	11.74	13.23	20.03	7.64	12.98	11.33	21.15	27.43	39.54	21.01	23.99	32.89	5.14	11.63	17.97	11.12	13.73	12.28	7.36	9.67	11.09	11.42	12.97	30.92
		ARC	12.05	18.66	24.34	10.16	10.77	12.56	14.53	27.50	38.29	17.94	23.31	30.12	7.03	12.58	16.48	9.60	10.23	13.64	7.27	8.36	10.21	13.72	15.55	26.38
	OTTA	ADCSD	10.26	16.02	26.09	11.72	12.32	13.38	17.76	28.27	37.89	20.45	23.68	31.01	10.65	11.23	17.27	10.85	11.11	13.55	8.63	8.87	11.79	15.63	16.88	28.73
		LCoTTA	10.36	13.27	23.49	11.56	10.07	12.98	14.43	27.17	38.43	16.49	27.77	30.98	10.62	11.17	17.13	11.85	11.13	13.74	7.68	9.89	11.75	14.47	13.67	29.40
		DP-DGAD (OTTA)	10.46	18.38	23.05	12.49	13.17	13.78	19.99	29.63	36.25	25.83	27.11	33.77	9.52	10.06	14.24	10.79	12.33	18.64	7.65	10.86	10.14	13.27	15.20	27.08
		OTTA-DGAD	12.38	23.32	31.68	17.67	16.08	15.30	23.95	36.47	40.28	30.47	30.74	40.84	11.84	12.97	18.18	12.80	15.07	23.56	9.67	12.88	16.07	17.52	20.47	32.37

the adaptation parameters; no cross chunk recovery, prototype update, or graph-specific replay is used.

To evaluate models under continually evolving target distributions, we adopt an online test-time adaptation (OTTA) setting on chunked target dynamic graphs. Specifically, 1) the model is first pre-trained on multiple labeled source datasets and then adapted to an unlabeled target dataset. To comply with the OTTA setting, the source-pretrained backbone is frozen during target-time adaptation, and only lightweight adaptation modules are allowed to be updated. 2) Each target dynamic graph is chronologically divided into temporally ordered chunks, where each chunk contains a fixed number of target edges, here we set as 128. For each incoming chunk, the model first performs inference on the current chunk, and then updates using only the current chunk, without accessing any future information. 3) This test-then-adapt process is repeated sequentially over the whole target stream, and the final performance is reported by averaging over all target chunks.

C. Experimental Settings

All methods undergo pretraining on two source datasets, Wikipedia and MOOC [13], chosen because they have available ground truth anomalies. Parameters for ego-graph extraction, GNN, and the transformer are aligned with those of GeneralDyG [14]. The number of confident detections for pseudo-labeling, N_{con} , is 10% of the chunk size. $\mathcal{M}_{c,max}$ is set to be 10% of the bigger training source dataset, MOOC. The momentum α is set to 0.9. The value of p is varied at 1%, 5%, and 10%. The loss ratios λ_A and λ_{BCE} are set to 0.1 and 0.9, respectively, while λ_d and λ_e are set to 0.3 and 0.7. λ_c is set to 0.5.

D. Experiments Results

In this section, we evaluate OTTA-DGAD's performance against three types of baselines, normal DGADs without

specifically designed generalization module, generalists with specifically designed module and OTTA models that specifically designed for OTTA settings. To the end, we have following observations: (1) OTTA-DGAD consistently perform better than all other baselines. DP-DGAD, can still have good performance, but sometimes fall back behind ADCSD and LCoTTA that have specific OTTA models designed, making them more good at adapting online. Although generalists generally perform well, they cannot effectively incorporate domain specific patterns, resulting in lower performance than OTTA models. DGADs have the worse performance, mostly due to the fact that they fall short in generalization. (2) Compared with the table in DP-DGAD, most models suffer from performance degradation, DGADs suffer most due to the continuously emerging new chunks to make model overfit. Generalist are relatively stables as aforementioned that they have generalist modules. DP-DGAD also have drop since there confident detection models overly rely on the entropy loss, not to mention the lose of information overchunks. (3) We take 10% anomaly ratio TAX51 data that have most edge numbers for better visualization chunk wise performance and the coefficient of variation over chunk and domains for stability analysis. As shown in Fig. 3(a) and (d), OTTA-DGAD consistently outperforms the other baselines. Also from Fig. 3 (b),(c),(f) and (e) that OTTA-DGAD is generally the most stable one across chunks and domains, while DP-DGAD and the other two OTTA baselines are among the next most stable methods.

In summary, from the above results, OTTA-DGAD has shown is state-of-art online performance as well as stability.

E. Ablation Study

In this section, we conduct experiments by removing key components in OTTA-DGAD to study their effectiveness: (1) **w/o DPA** replaces the dynamic prototype-based anomaly

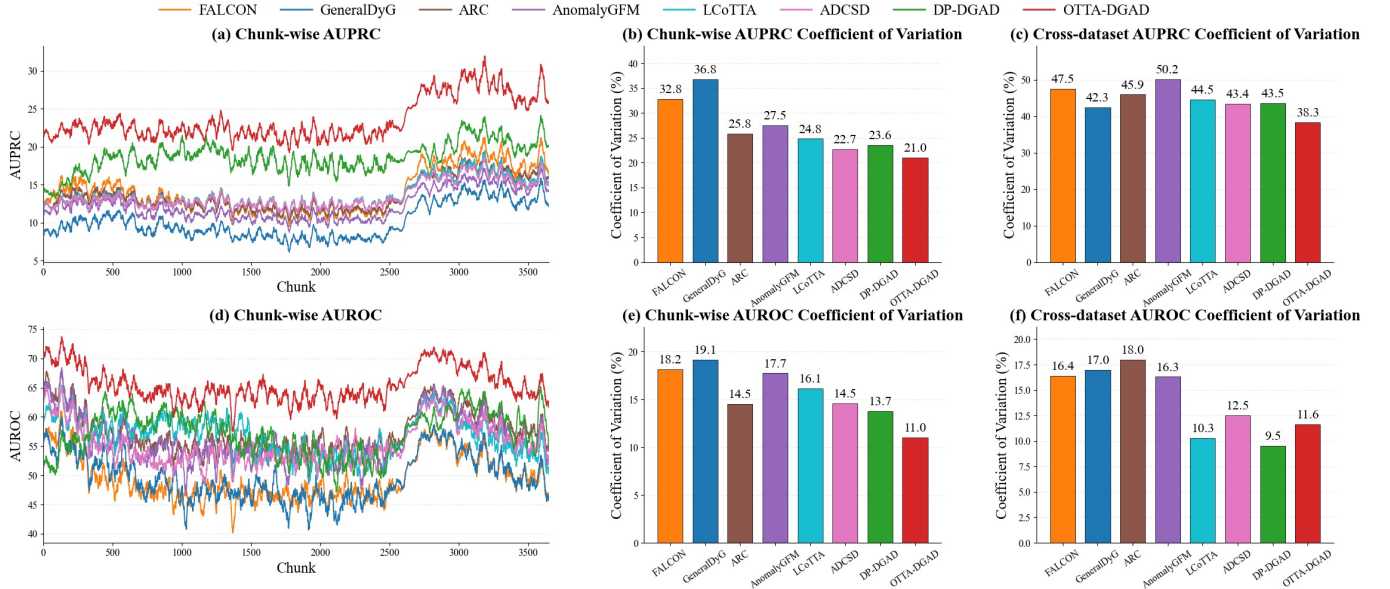


Fig. 3. Visualization of chunk-wise metric and coefficient of variation over domain/chunks on TAX51 10% anomaly ratio.

TABLE III
ABLATION RESULTS OF DIFFERENT VARIANTS OF OTTA-DGAD.

Metric	Variants	Dataset							
		AS-Topology	Bitcoin-Alpha	Email-DNC	UCI Messages	Bitcoin-OTC	TAX51	DBLP	Synthetic-Hijack
AUROC	w/o DPA	53.64	62.21	66.77	58.90	70.53	44.11	49.06	60.12
	w/o DPAD	40.10	45.03	47.82	50.41	54.74	48.09	45.09	47.22
	w/o CDGA	63.34	69.98	72.50	60.19	74.57	60.13	55.58	70.26
	w/o Wiki	60.01	53.16	67.21	52.83	48.04	46.32	50.28	67.97
	w/o MOOC	55.86	60.07	53.15	60.37	62.94	56.36	46.12	58.81
	w/o CCCE	63.38	62.53	67.04	61.13	68.76	54.25	60.84	68.71
	w/o SGPU	66.47	67.81	70.26	63.65	69.11	60.08	60.30	73.49
	OTTA-DGAD	76.99	75.47	81.77	69.75	78.61	66.06	63.25	83.72
AUPRC	w/o DPA	19.73	12.26	32.15	22.36	16.14	15.45	9.01	28.25
	w/o DPAD	13.33	12.41	10.59	17.07	11.49	10.23	11.59	17.66
	w/o CDGA	16.86	10.05	30.94	30.92	15.18	17.54	8.21	27.07
	w/o Wiki	20.73	10.25	26.34	20.58	16.09	10.21	12.91	25.18
	w/o MOOC	17.22	12.13	20.69	32.37	12.19	16.30	10.06	23.53
	w/o CCCE	25.83	12.77	34.59	30.07	15.29	19.01	12.88	25.67
	w/o SGPU	27.15	11.56	31.09	30.14	12.87	20.55	10.11	26.24
	OTTA-DGAD	31.68	15.30	40.28	40.84	18.18	23.56	16.07	32.37

scoring with a normal binary classifier. (2) **w/o DPAD** sets λ_e to 0 and λ_d to 1, thereby removing the domain adaptation module from the domain adaptive anomaly scoring in OTTA-DGAD. (3) **w/o CDGA** removes the confident detection guided memory buffer update on the target dataset, using the model and buffer pretrained on source datasets. (4) **w/o Wiki** pretrains OTTA-DGAD on MOOC only. (5) **w/o MOOC** pretrains OTTA-DGAD on Wiki only. (6) **w/o CCCE** removes the cross chunk context enrichment. (7) **w/o SGPU** removes the structural guided prototype update, using entropy based update only.

From the above Table. III, OTTA-DGAD outperforms all ablated versions. Additionally, we have several observations: (1) The adaptation modules enable the model to better adapt to new domains while dynamic prototype updates allow it to adapt to individual chunks within each domain. Removing such module will lead to performance degradation. (2) Cross

chunk information is highly important, because it helps the model generalize across chunks and prevents it from forgetting important historical pattern. (3) Removing the confident detection will lead to fail of capturing new patterns in unlabeled new domain. However, entropy based detection is not enough as w/o SGPU also suffer from performance dropping compared with the OTTA-DGAD.

F. Hyperparameter Analysis

To analyze OTTA-DGAD's sensitivity to hyperparameters, we examine four key parameters: the dynamic prototype memory buffer size $\mathcal{M}$, the momentum α , top confident detection number N_{con} (Here we set it as a different proportion to the full dataset size) and the chunks size every time processed. We vary these values and observe their impact on model performance in Fig. 4. Our findings are as follows. (1) **Increasing the memory buffer size generally improve**

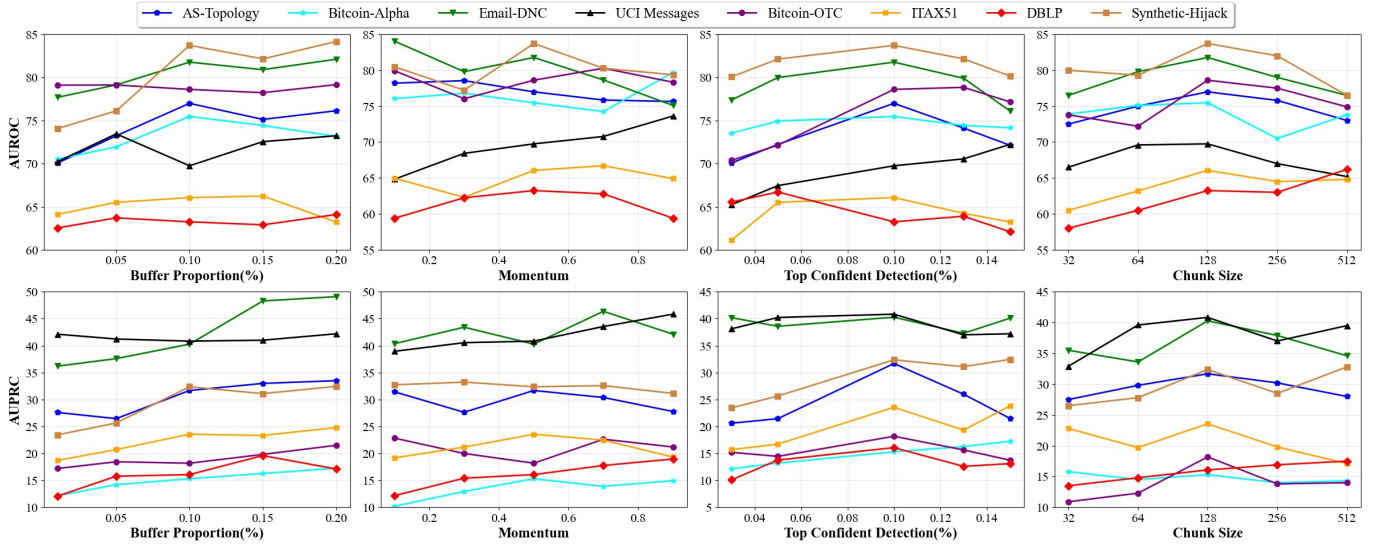


Fig. 4. Impact of dynamic prototype memory buffer size $\mathcal{M}$, momentum α , confident detection N_{con} and chunk size on the OTTA-DGAD performance.

performance. This is because a larger memory buffer contains more prototype pairs, thus covering broader aspects of the distribution of abnormal patterns. However, we observe an intriguing pattern on some datasets, that Larger buffer sizes actually degrade performance. This occurs because oversized buffers may include irrelevant prototype pairs, which are neither similar to nor distinct enough from the new domains. They introduce bias and interfere with anomaly scoring, which relies on buffer prototypes to extract distributions. **(2) Using the top 10% most confident detections appears to be the optimal setting.** On most datasets, performance improves as the top confident detection number increases. However, performance begins to drop when this percentage exceeds 10%. This decline occurs because including more than the top 10% of detections reduces reliability. These less confident detections have high entropy and introduce additional noise into the pseudo-labeling process. **(3) OTTA-DGAD remains stable under different momentum.** We vary the momentum value and find OTTA-DGAD remains stable across all datasets with no clear trend as momentum increases. This stability arises from the fact that the memory buffer already captures sufficient similar abnormal and normal patterns through the selection score s_r constraint. As a result, the extracted distributions remain stable regardless of α 's value. **(4) Chunk size equal to 128 appears to be the optimal setting.** This results is reason from that big chunk size such as 256 and 512 may lead to overfit to certain chunks while small chunk size may lead to unbalanced issues.

VI. VISUALIZATION

In this section, we visualize the representation enriched by cross chunk context and also the comparison between entropy only and structural support added pseduo-labeling.

It can be observed from left part of Fig. 5 that 97.6% of normal edges and 86.1% of anomalous edges move closer to the true distribution after enrichment and more nodes fall in the bottom right corner of the no change diagonal line.

Combining structural support with entropy-based confidence corrects cases where a prediction is confidently wrong but structurally inconsistent with its neighborhood. As shown in Fig. 5 right part, among the 1,654 chunks where the two criteria diverge, structural supported model wins 66.6% of the time, and McNemar's test confirms this advantage is highly significant, rather than attributable to chance. The large number of ties is expected since many chunks have limited anomaly samples.

VII. CONCLUSION

In this paper, we propose a generalist DGAD detector that achieves robust performance when target data arrive as online unlabeled chunks. Specifically, we aim to capture the evolving anomalous patterns that are both domain agnostic and domain specific, thereby achieving better generalizability than existing methods. To this end, we introduce OTTA-DGAD, a generalist detector that leverages dynamic prototypes extracted from temporal ego-graphs to model evolving anomalies. We further update the memory buffer using confident pseudo-labels, enabling effective adaptation to an unlabeled target domain. A Cross chunk context enrichment module is added to preserve historical evolving patterns. Extensive experiments across 10 real-world datasets from diverse domains validate the effectiveness of OTTA-DGAD.

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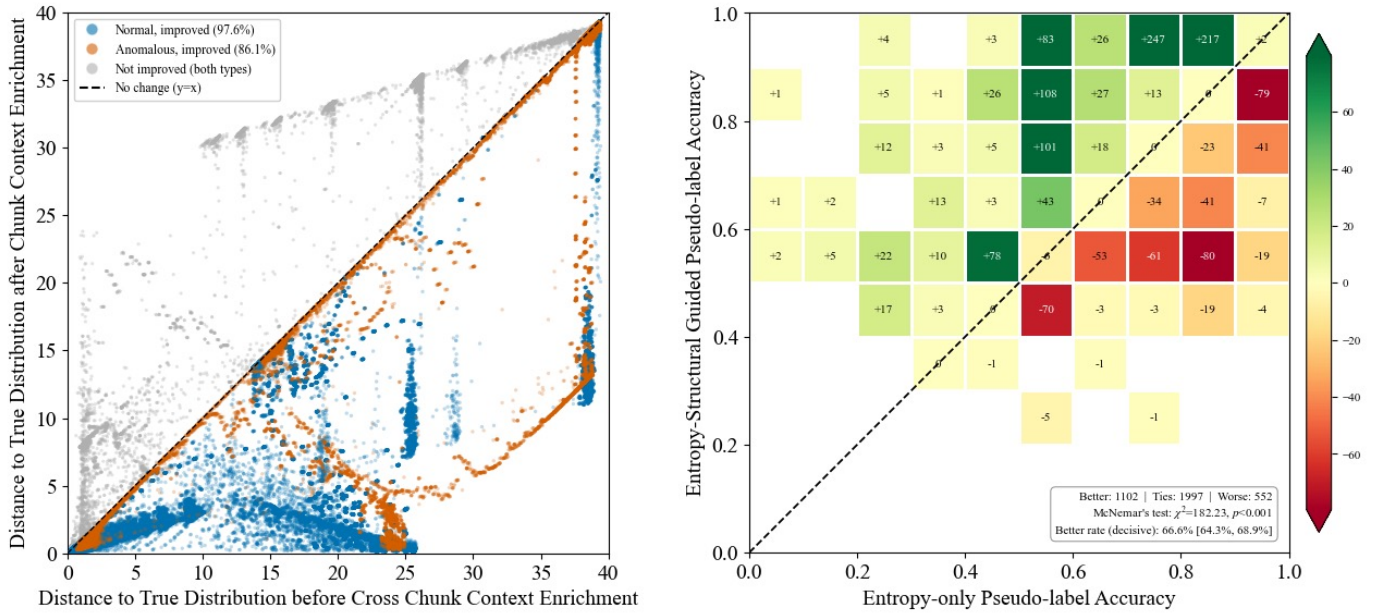


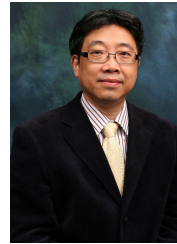
Fig. 5. Effect of Cross-Chunk Context Enrichment and Structural-Guided Pseudo-labeling accuracy.

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